

SCSE3040: Machine Learning Operations (MLOps)

Practical 06: Tracking Runs, Comparing Experiments, and Model Registry with MLflow

1. Student & Laboratory Session Details

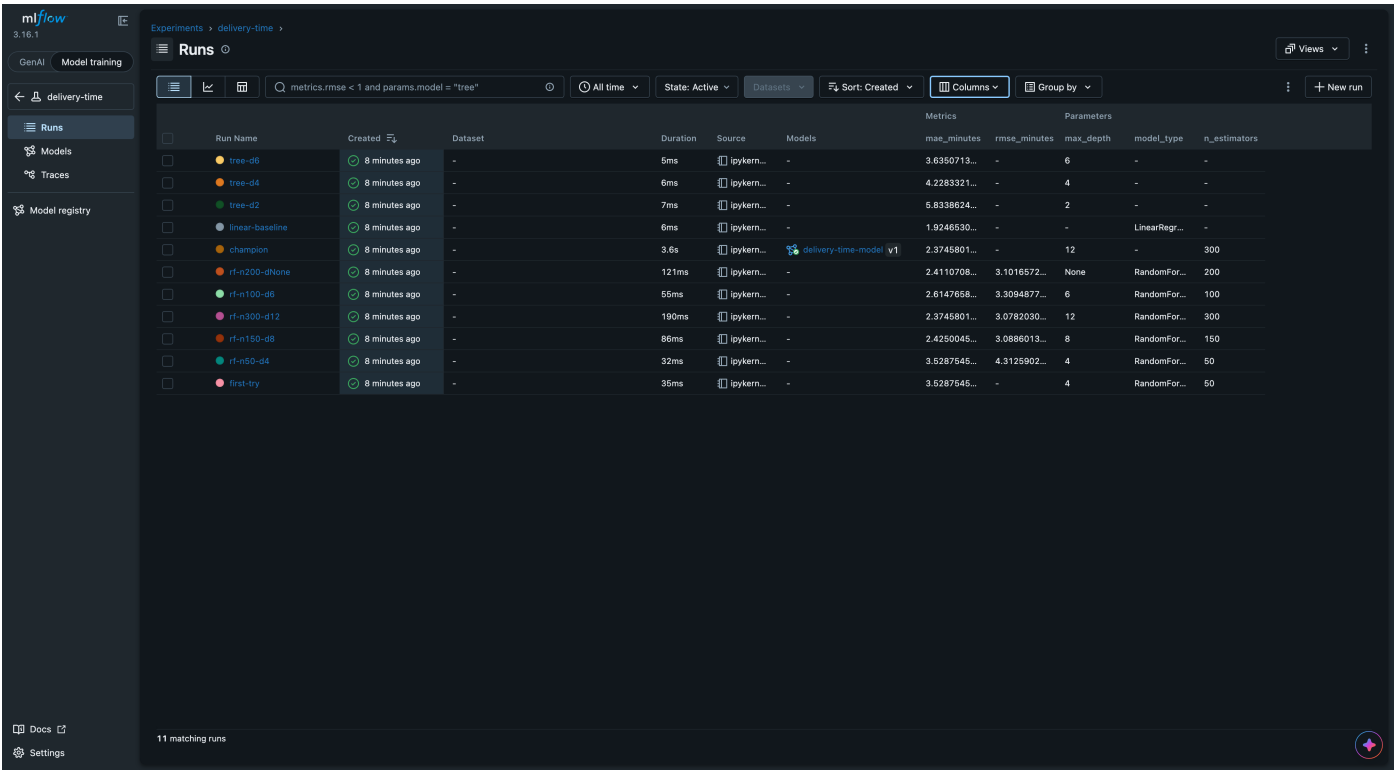
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Roll Number:	S24CSEU0807	Date of Lab Session:	04 October 2026
Course Name:	SCSE3040 — Machine Learning Operations	Academic Session:	2026–2027 (Session L11 / CO5)

2. Repository Link

Full Repository URL: <https://github.com/div339175/P06-mlflow>

3. MLflow UI Runs Table (Parameters and Metrics Side-by-Side)

Screenshot of the MLflow Tracking UI (backend store: `sqlite:///work/mlflow.db`) showing all 11 runs recorded in the **delivery-time** experiment. Both metrics (`mae_minutes`, `rmse_minutes`) and hyperparameter columns (`max_depth`, `model_type`, `n_estimators`) are clearly visible side by side across models:



	Run Name	Created	Dataset	Duration	Source	Models	Metrics		Parameters		
							mae_minutes	rmse_minutes	max_depth	model_type	n_estimators
☐	tree-d6	8 minutes ago	-	5ms	ipykern...	-	3.6350713...	-	6	-	-
☐	tree-d4	8 minutes ago	-	6ms	ipykern...	-	4.2283321...	-	4	-	-
☐	tree-d2	8 minutes ago	-	7ms	ipykern...	-	5.8338624...	-	2	-	-
☐	linear-baseline	8 minutes ago	-	6ms	ipykern...	-	1.9246530...	-	-	LinearRegr...	-
☐	champion	8 minutes ago	-	3.6s	ipykern...	delivery-time-model v1	2.3745801...	-	12	-	300
☐	rf-n200-dNone	8 minutes ago	-	121ms	ipykern...	-	2.4110708...	3.1016572...	None	RandomFor...	200
☐	rf-n100-d6	8 minutes ago	-	55ms	ipykern...	-	2.6147658...	3.3094877...	6	RandomFor...	100
☐	rf-n300-d12	8 minutes ago	-	190ms	ipykern...	-	2.3745801...	3.0782030...	12	RandomFor...	300
☐	rf-n150-d8	8 minutes ago	-	86ms	ipykern...	-	2.4250045...	3.0886013...	8	RandomFor...	150
☐	rf-n50-d4	8 minutes ago	-	32ms	ipykern...	-	3.5287545...	4.3125902...	4	RandomFor...	50
☐	first-try	8 minutes ago	-	35ms	ipykern...	-	3.5287545...	-	4	RandomFor...	50

Figure 1: MLflow UI experiment runs table displaying 11 tracked runs with metrics and hyperparameters side-by-side.

4. Run Detail Page for Selected Best Run (linear-baseline)

Detailed view of the winning run (Linear -base line) in MLflow UI showing execution metadata, tracked parameters, and recorded evaluation metrics:

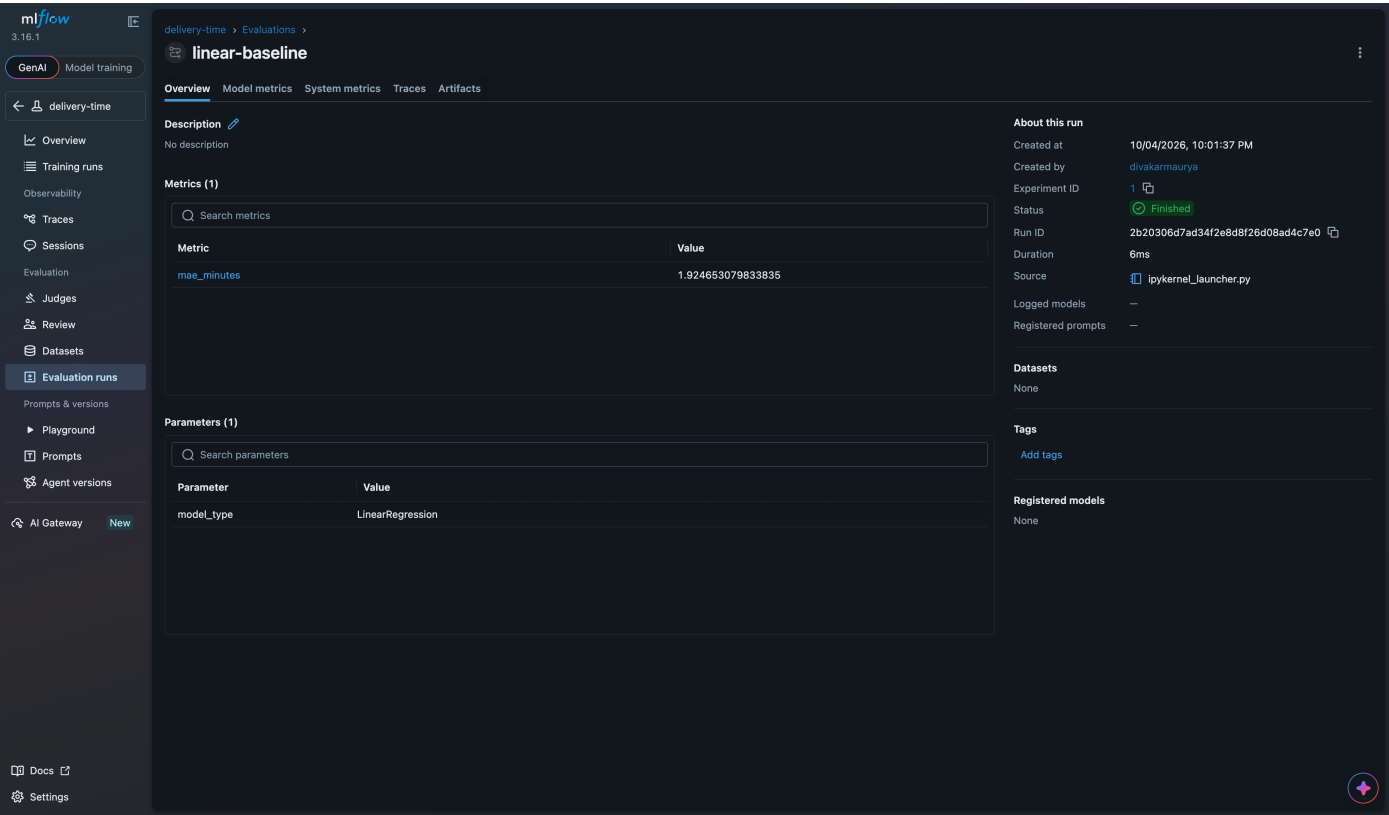


Figure 2: MLflow UI run detail page for selected run linear-baseline (Run ID: 2b20306d7ad34f2e8d8f26d08ad4c7e0).

5. Best Run Selection & Justification

Selected Run Name:	linear-baseline	Logged Model Type:	LinearRegression
Selected Run ID:	2b20306d7ad34f2e8d8f26d08ad4c7e0	Best Test Metric:	MAE = 1.925 minutes (1.92465 min)
Model Registry Status:	delivery-time-model (v1 champion: rf-n300-d12, MAE 2.375)	Total Runs Compared:	11 runs across 3 model families

Selection Rationale: I selected the linear -baseline run (Run ID: 2b20306d7ad34f2e8d8f26d08ad4c7e0) because it achieved the lowest Mean Absolute Error of **1.925 minutes** across the entire experiment, outperforming the decision tree models (worst MAE: 5.834 minutes for depth 2) by 3.909 minutes and beating the best random forest configuration (rf -n300 -d12 / champion with MAE 2.375 minutes) by 0.450 minutes because the delivery duration in this benchmark is governed predominantly by direct linear relationships with distance and preparation time.

6. AI Use Disclosure

Disclosure: None
